Biochemistry Cume
January 25, 2013

1. The template strand of a segment of double-helical DNA contains the sequence,

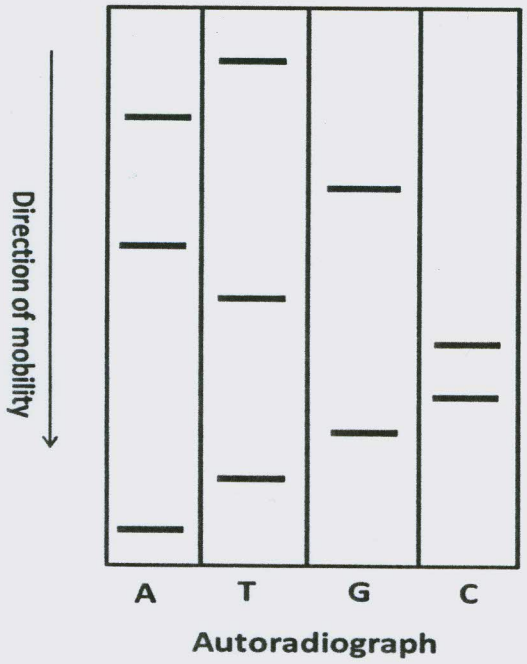
5' – CTTAACACCCCTGACTTCGCGCCGTCG – 3' (**18 points**)

A. What is the base sequence of the mRNA that can be transcribed from this strand?
B. What amino acid sequence could be coded by the mRNA starting from the 5'-end?
C. If the complementary strand of this DNA were transcribed and translated, would the resulting amino acid sequence be the same as in part-**B**?

2. Calculate the length of a double stranded B-DNA molecule of molecular weight 1.5×10^7 Daltons. (Given: the average molecular weight of a complementary deoxynucleotide pair is 618 Daltons and the distance between each base-pair in B-DNA is 3.4 Å). (**25 points**)

3. The percentage of Adenine in *E.coli* DNA is thrice the percentage Guanine. Calculate the percentage concentrations of Adenine, Guanine, Thymine, and Cytosine. (**20 points**)

4. A. Given below is an autoradiograph of a sequencing gel of a segment of B-DNA that was treated according to the chain terminator method of DNA sequencing. What is the sequence of the DNA? (**24 points**)



B. Calculate the length of the B-DNA sequence identified in Part-**A**? **(13 points)**

Genetic code

First	U	C	A	G	Last
U	Phe	Ser	Tyr	Cys	**U**
	Phe	Ser	Tyr	Cys	**C**
	Leu	Ser	**Stop**	**Stop**	**A**
	Leu	Ser	**Stop**	Trp	**G**
C	Leu	Pro	His	Arg	**U**
	Leu	Pro	His	Arg	**C**
	Leu	Pro	Gln	Arg	**A**
	Leu	Pro	Gln	Arg	**G**
A	Ile	Thr	Asn	Ser	**U**
	Ile	Thr	Asn	Ser	**C**
	Ile	Thr	Lys	Arg	**A**
	Met	Thr	Lys	Arg	**G**
G	Val	Ala	Asp	Gly	**U**
	Val	Ala	Asp	Gly	**C**
	Val	Ala	Glu	Gly	**A**
	Val	Ala	Glu	Gly	**G**